

| Board Technology | Min Trace Width | 0.2mm |
|---|---|---|
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| Silkscreen | Color | White |
| | | |
| Material | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| Copper surface | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| Solder mask | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| Drilling | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| Manufacturing | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| Testing | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| Modifications | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

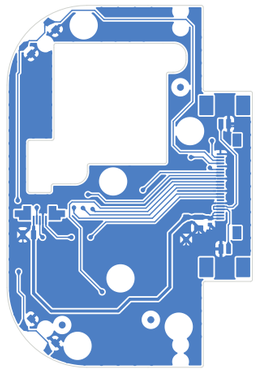
Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
| | |
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| Designed by: Jean–Francois Bilodeau<br>P.Eng #6022173<br>203–1690 rue des quatre–saisons<br>Sherbrooke, Quebec, J1E0N2<br>KiCad Rev 10.0 | | Logo |
|---|---|---|
| | | Sheet Size: A3 |
| Project: ConJoy [Console Joystick] | Drawn by: Jean–Francois Bilodeau | Sheet: 1 of 10 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Status: In Review |
| Date: 2026–11–01 | BOM: Conjoy_1.00–0A_BOM | Current Revision: 1.00–0A |



| Board Technology | Min Trace Width | 0.2mm |
|---|---|---|
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| **Silkscreen** | Color | White |
| | | |
| **Material** | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| **Copper surface** | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| **Solder mask** | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| **Drilling** | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| **Manufacturing** | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| **Testing** | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| **Modifications** | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

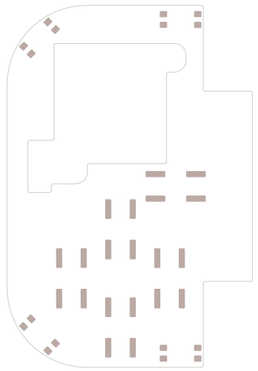
Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
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|---|---|---|
| Designed by: Jean–Francois Bilodeau<br>P.Eng #6022173<br>203–1690 rue des quatre–saisons<br>Sherbrooke, Quebec, J1E0N2<br>KiCad Rev 10.0 | | Logo<br>Sheet Size: A3 |
| Project: ConJoy [Console Joystick] | Drawn by: Jean–Francois Bilodeau | Sheet: 2 of 10 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Status: In Review |
| Date: 2026–11–01 | BOM: Conjoy_1.00–0A_BOM | Current Revision: 1.00–0A |



| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026-11-01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad-PTH-drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

Total plated holes count 136

Drill file 'ConJoy_KiCad-NPTH-drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| | | |
|---|---|---|
| **Board Technology** | Min Trace Width | 0.2mm |
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| **Silkscreen** | Color | White |
| | | |
| **Material** | Environmental | Lead-free and capable of meeting a lead-free assembly process |
| | Material | FR-4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL-94V0 and UL-796 |
| | Bow & Twist | Class A |
| | | |
| **Copper surface** | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC-A-600 latest revision |
| | | |
| **Solder mask** | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| **Drilling** | Hole tolerance | IPC-A-600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| **Manufacturing** | Compliance | IPC-6012 Class 2; IPC-600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| **Testing** | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| **Modifications** | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

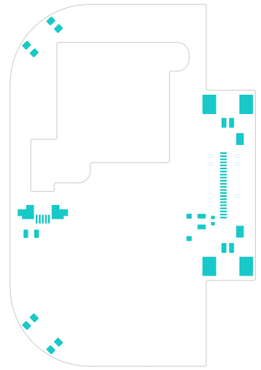
| Revision | Change Order |
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Designed by: Jean-Francois Bilodeau
P.Eng #6022173
203-1690 rue des quatre-saisons
Sherbrooke, Quebec, J1E0N2

KiCad Rev 10.0

Logo

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|---|---|---|
| Project: ConJoy [Console Joystick] | Drawn by: Jean-Francois Bilodeau | Sheet Size: A3 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Sheet: 3 of 10 |
| Date: 2026-11-01 | BOM: Conjoy_1.00-0A_BOM | Status: In Review |
| | | Current Revision: 1.00-0A |



| | | |
|---|---|---|
| **Board Technology** | Min Trace Width | 0.2mm |
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| **Silkscreen** | Color | White |
| | | |
| **Material** | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| **Copper surface** | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| **Solder mask** | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| **Drilling** | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| **Manufacturing** | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| **Testing** | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| **Modifications** | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
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|---|---|---|
| *Designed by: Jean–Francois Bilodeau*<br>*P.Eng #6022173*<br>*203–1690 rue des quatre–saisons*<br>*Sherbrooke, Quebec, J1E0N2*<br>*KiCad Rev 10.0* | | Logo<br>*Sheet Size: A3* |
| *Project: ConJoy [Console Joystick]* | *Drawn by: Jean–Francois Bilodeau* | *Sheet: 4 of 10* |
| *Sheet Title: ConJoy_KiCad.kicad_pcb* | *Approved by: Not applicable* | *Status: In Review* |
| *Date: 2026–11–01* | *BOM: Conjoy_1.00–0A_BOM* | *Current Revision: 1.00–0A* |

| Board Technology | Min Trace Width | 0.2mm |
|---|---|---|
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| Silkscreen | Color | White |
| | | |
| Material | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| Copper surface | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| Solder mask | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| Drilling | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| Manufacturing | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| Testing | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| Modifications | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

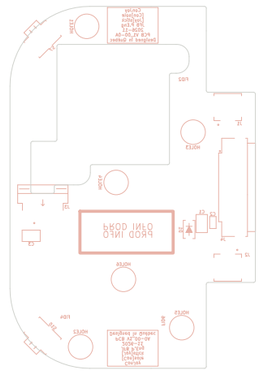
Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
| | |
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| Designed by: Jean–Francois Bilodeau<br>P.Eng #6022173<br>203–1690 rue des quatre–saisons<br>Sherbrooke, Quebec, J1E0N2<br>KiCad Rev 10.0 | | Logo |
|---|---|---|
| | | Sheet Size: A3 |
| Project: ConJoy [Console Joystick] | Drawn by: Jean–Francois Bilodeau | Sheet: 5 of 10 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Status: In Review |
| Date: 2026–11–01 | BOM: Conjoy_1.00–0A_BOM | Current Revision: 1.00–0A |


HOLE
HOLE
HOLE

| Board Technology | Min Trace Width | 0.2mm |
|---|---|---|
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| Silkscreen | Color | White |
| | | |
| Material | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| Copper surface | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| Solder mask | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| Drilling | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| Manufacturing | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| Testing | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| Modifications | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026-11-01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad-PTH-drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

Total plated holes count 136

Drill file 'ConJoy_KiCad-NPTH-drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
| | |
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| Designed by: Jean–Francois Bilodeau<br>P.Eng #6022173<br>203–1690 rue des quatre–saisons<br>Sherbrooke, Quebec, J1E0N2<br>KiCad Rev 10.0 | | Logo |
|---|---|---|
| | | Sheet Size: A3 |
| Project: ConJoy [Console Joystick] | Drawn by: Jean–Francois Bilodeau | Sheet: 7 of 10 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Status: In Review |
| Date: 2026–11–01 | BOM: Conjoy_1.00–0A_BOM | Current Revision: 1.00–0A |

| Board Technology | Min Trace Width | 0.2mm |
|---|---|---|
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| Silkscreen | Color | White |
| | | |
| Material | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| Copper surface | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| Solder mask | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| Drilling | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| Manufacturing | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| Testing | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| Modifications | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
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| Designed by: Jean–Francois Bilodeau<br>P.Eng #6022173<br>203–1690 rue des quatre–saisons<br>Sherbrooke, Quebec, J1E0N2<br>KiCad Rev 10.0 | | Logo |
|---|---|---|
| | | Sheet Size: A3 |
| Project: ConJoy [Console Joystick] | Drawn by: Jean–Francois Bilodeau | Sheet: 8 of 10 |
| Sheet Title: ConJoy_KiCad.kicad_pcb | Approved by: Not applicable | Status: In Review |
| Date: 2026–11–01 | BOM: Conjoy_1.00–0A_BOM | Current Revision: 1.00–0A |

| | | |
|---|---|---|
| **Board Technology** | Min Trace Width | 0.2mm |
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| **Silkscreen** | Color | White |
| | | |
| **Material** | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| **Copper surface** | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| **Solder mask** | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| **Drilling** | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| **Manufacturing** | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| **Testing** | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| **Modifications** | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
==========
T1 0.300mm 0.0118" (136 holes))

Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
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|---|---|---|
| *Designed by: Jean–Francois Bilodeau*<br>*P.Eng #6022173*<br>*203–1690 rue des quatre–saisons*<br>*Sherbrooke, Quebec, J1E0N2*<br>*KiCad Rev 10.0* | | Logo |
| | | *Sheet Size: A3* |
| *Project: ConJoy [Console Joystick]* | *Drawn by: Jean–Francois Bilodeau* | *Sheet: 9 of 10* |
| *Sheet Title: ConJoy_KiCad.kicad_pcb* | *Approved by: Not applicable* | *Status: In Review* |
| *Date: 2026–11–01* | *BOM: Conjoy_1.00–0A_BOM* | *Current Revision: 1.00–0A* |

| | | |
|---|---|---|
| **Board Technology** | Min Trace Width | 0.2mm |
| | Min via size | 0.3mm – 0.6mm |
| | Min spacing | 0.25mm |
| | Layer count | 2x |
| | Impedance Matching | No |
| | Teardrops | By design |
| | Dimensions (L x W x H) | 40.00mm x 59.00mm |
| | | |
| **Silkscreen** | Color | White |
| | | |
| **Material** | Environmental | Lead–free and capable of meeting a lead–free assembly process |
| | Material | FR–4 >= TG150 or equivalent |
| | Finished thickness | 1.6mm ±10% |
| | Compliance | Must meet UL–94V0 and UL–796 |
| | Bow & Twist | Class A |
| | | |
| **Copper surface** | Finish process | ENIG >= 1u thickness |
| | Finished plating | Must meet IPC–A–600 latest revision |
| | | |
| **Solder mask** | Finish | Type SMOBC |
| | Thickness | 0.4mils – 1.2mils |
| | Color | As specified in order |
| | | |
| **Drilling** | Hole tolerance | IPC–A–600 latest revision Class 2 |
| | Hole size and count | See drilling table |
| | PTH and NPTH | Separate drilling files |
| | | |
| **Manufacturing** | Compliance | IPC–6012 Class 2; IPC–600G Class 2 |
| | Material | All materials must be RoHS |
| | UL number | UL EXXXXX file number and UL flammability compliance must be added on TOP SILK |
| | X out allowed | 1x OUT allowed |
| | | |
| **Testing** | Electrical test | All PCB must be electrically tested |
| | Test coupon | Test coupon must be added on each panel and coupon be returned with test report on each production lot |
| | | |
| **Modifications** | Warning | Customer must be advised of any modifications to gerber files or derogation to any of the requirements |

| Layer Name | Type | Material | Thickness | Color | Epsilon R | Loss Tangent |
|---|---|---|---|---|---|---|
| F.Silkscreen | Top silk screen | Not specified | Not specified | Not specified | 1 | 0 |
| F.Paste | Top solder paste | | Not specified | | 1 | 0 |
| F.Mask | Top solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| F.Cu | Copper | | 0.035 mm | | 1 | 0 |
| Dielectric 1 | Core | FR4 | 1.51 mm | Not specified | 4.5 | 0.02 |
| B.Cu | Copper | | 0.035 mm | | 1 | 0 |
| B.Mask | Bottom solder mask | Not specified | 0.01 mm | Not specified | 3.3 | 0 |
| B.Paste | Bottom solder paste | | Not specified | | 1 | 0 |
| B.Silkscreen | Bottom silk screen | Not specified | Not specified | Not specified | 1 | 0 |

Drill report for ConJoy_KiCad.kicad_pcb
Created on 2026–11–01T17:19:57

Copper Layer Stackup:
==========
L1 : F.Cu front
L2 : B.Cu back

Drill file 'ConJoy_KiCad–PTH–drl.gbr' contains
plated through holes:
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Total plated holes count 136

Drill file 'ConJoy_KiCad–NPTH–drl.gbr' contains
unplated through holes:
==========
T1 0.750mm 0.0295" (12 holes))
T2 2.600mm 0.1024" (6 holes))

Total unplated holes count 18

| Revision | Change Order |
|---|---|
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|---|---|---|
| *Designed by: Jean–Francois Bilodeau*<br>*P.Eng #6022173*<br>*203–1690 rue des quatre–saisons*<br>*Sherbrooke, Quebec, J1E0N2*<br>*KiCad Rev 10.0* | | Logo<br>*Sheet Size: A3* |
| *Project: ConJoy [Console Joystick]* | *Drawn by: Jean–Francois Bilodeau* | *Sheet: 10 of 10* |
| *Sheet Title: ConJoy_KiCad.kicad_pcb* | *Approved by: Not applicable* | *Status: In Review* |
| *Date: 2026–11–01* | *BOM: Conjoy_1.00–0A_BOM* | *Current Revision: 1.00–0A* |